# SuperMagnifierLV

## User Guide

v2026.05.09.0000

Windows Magnification API • 1.25×–20× • 5 Follow Modes • 9 Colour Filters

Windows 10/11 • .NET 10 • WPF/MVVM

# 1. Overview

SuperMagnifierLV is a hardware-accelerated screen magnifier built on the Windows Magnification API. It provides a resizable floating Lens window that can follow the mouse cursor, keyboard caret, or specific application windows, with 9 real-time colour filters, 5 overlay types, an integrated colour picker, per-application automatic preset switching, and per-monitor zoom memory.

The Magnification API renders the lens contents via DWM composition, ensuring low latency and no CPU-side pixel copying.

# 2. System Requirements

| Component | Requirement |
|---|---|
| OS | Windows 10 v1903 or Windows 11 |
| Runtime | .NET 10 Desktop Runtime (x64) |
| RAM | 64 MB minimum |
| GPU | DirectX 9 compatible (for DWM compositing) |
| Display | Any resolution; DPI-aware per-monitor v2 |
| Permissions | Standard user — no elevation required |

# 3. Installation

Run the SuperMagnifierLV installer or use the portable ZIP. Settings stored in %APPDATA%\SuperMagnifierLV\settings.json. Per-app presets stored in %APPDATA%\SuperMagnifierLV\app_presets.json.

# 4. Interface Tour

## 4.1 Control Panel

| Zone | Purpose |
|---|---|
| 1 Identity/Scale | App name, version, DPI scale selector |
| 2 NavRow | Zoom \| Follow \| Filters \| Overlays \| Picker \| Presets \| Settings |
| 3 Toggle/Preview | Lens on/off toggle + miniature lens preview |
| 4 Main / Sliders | Active panel — zoom sliders, step size, filter dropdowns, overlay toggles |
| 5 Presets 1–5 / Save | Preset slots; resize grip bottom-right |

## 4.2 Lens Window

The Lens window is a borderless, always-on-top floating panel. Drag its title bar to reposition. Drag the bottom-right resize grip to resize. Double-click the title bar to toggle between lens and full-screen magnification modes.

# 5. Zoom Controls

| Control | Range | Notes |
|---|---|---|
| Zoom Level slider | 1.25× – 20× | Smooth drag; also Ctrl+Scroll anywhere on desktop |
| Step Size | 0.25× / 0.5× / 1× / 2× / 5× / 10× | Increment for hotkey zoom in/out |
| Zoom In hotkey | Ctrl+= (default) | Increases by current step size |
| Zoom Out hotkey | Ctrl+- (default) | Decreases by current step size |
| Reset Zoom | Ctrl+0 | Returns to 1.0× (no magnification) |
| Per-Monitor Memory | Automatic | Zoom level saved per monitor on disconnect |

■ Ctrl+Scroll works globally without focus on the control panel.

# 6. Follow Modes

| Mode | Behaviour |
|---|---|
| Cursor | Lens centre tracks the mouse cursor in real time (default) |
| Caret | Lens tracks the text insertion caret in the focused application (uses UIA) |
| Snap-to-Word | Lens snaps to the word under the cursor with a configurable dwell delay |
| Edge Dead Zone | Cursor must approach within N px of lens edge before lens moves — reduces drift |
| Fixed | Lens stays at a pinned screen position; user pans manually with arrow keys |

■ Edge Dead Zone threshold is configurable in Settings → Follow → Dead Zone Radius (default 40 px).

# 7. Colour Filters

Apply a real-time colour filter to the magnified content. Filters process on the GPU via the Magnification API's colour matrix:

| Filter | Description |
|---|---|
| None | Default — no colour transformation |
| Greyscale | Removes saturation; useful for contrast assessment |
| Invert | Inverts all channels; classic dark-mode simulation |
| Invert Greyscale | Inverted then desaturated |
| Deuteranopia | Green-weak simulation (red-green colour blindness) |
| Protanopia | Red-weak simulation |
| Tritanopia | Blue-yellow colour blindness simulation |
| High Contrast | Boosts contrast ratio; pushes colours to extremes |

| | |
|---|---|
| Warm | Reduces blue channel; comfortable for low-light reading |

■ Colour filters apply only within the lens; the rest of the screen is unaffected.

# 8. Overlays

| Overlay | Description |
|---|---|
| Reading Ruler | Horizontal band across the lens; height and colour configurable; aids line tracking |
| Reticle | Centre crosshair within the lens (independent of ReticleOverlay app) |
| Cursor Highlighter | Coloured halo around the cursor position in the lens |
| Cursor Finder Pulse | Animated expanding ring on cursor; helps locate cursor after saccades |
| Focus Ring | Highlights the keyboard-focused UI element within the lens (uses UIA) |

■ Overlays are composited after the colour filter, so they appear in their true colours regardless of filter.

# 9. Colour Picker

Open NavRow → Picker to sample any pixel on screen. Click anywhere on the desktop while the picker is active to capture the colour at that point. The picker displays:

• Hex value (#RRGGBB) • RGB components • HSL components • WCAG contrast ratio against white and black backgrounds

History stores the last 10 sampled colours. Click a history swatch to restore it. Right-click to copy the hex value to clipboard.

# 10. Presets & Per-App Switching

Five preset slots store the complete magnifier state: zoom level, step size, follow mode, colour filter, all overlay settings, and lens size/position.

Per-App Auto-Presets: In Settings → App Presets, assign a preset to a process name. When that process becomes the foreground window, SuperMagnifierLV automatically loads the assigned preset. When focus leaves, the previous preset is restored.

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Example: notepad.exe → Preset 2 (Caret follow, 4×, Reading Ruler on)
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# 11. Hotkeys

| Hotkey | Action |
|---|---|
| Ctrl+= / Ctrl+- | Zoom in / out by current step size |
| Ctrl+0 | Reset zoom to 1× |
| Ctrl+Scroll (global) | Zoom in/out from anywhere on desktop |
| Ctrl+Alt+L | Toggle lens on/off |
| Ctrl+Alt+F | Cycle follow modes |

| | |
|---|---|
| Ctrl+Alt+C | Open colour picker (one-click sample mode) |
| Ctrl+Alt+1–5 | Load preset 1–5 |
| Ctrl+Alt+N / P | Next / previous preset |
| Arrow keys (Fixed mode) | Pan lens 10 px in direction |
| Shift+Arrow (Fixed mode) | Pan lens 1 px in direction |
| Ctrl+Alt+T | Cycle colour filter |
| Ctrl+Alt+R | Toggle reading ruler |
| Ctrl+Alt+H | Toggle cursor highlighter |
| Ctrl+Alt+D | Toggle cursor finder pulse |
| Alt+S | Open Settings |
| Alt+T | Cycle theme |
| Win+Ctrl+M | Announce current zoom level via TTS |
| Alt+H | Hide control panel to tray |
| Ctrl+Alt+M | Toggle full-screen magnification mode |
| Esc | Exit picker / cancel current action |

# 12. Settings Reference

| Setting | Type | Default | Description |
|---|---|---|---|
| ZoomLevel | Float | 2.0 | Current magnification (1.25–20) |
| ZoomStep | Enum | 1× | Step size for hotkey zoom |
| FollowMode | Enum | Cursor | Cursor/Caret/SnapToWord/EdgeDeadZone/Fixed |
| DeadZoneRadius | Int | 40 | px threshold for EdgeDeadZone mode |
| ColourFilter | Enum | None | One of 9 filter names |
| LensWidth / LensHeight | Int | 400/300 | Initial lens dimensions (px) |
| ShowReadingRuler | Bool | false | Enable reading ruler overlay |
| RulerHeight | Int | 24 | Ruler band height in px |
| RulerColour | String | #ffd70040 | Ruler ARGB colour |
| ShowReticle | Bool | false | Lens crosshair overlay |
| ShowCursorHighlight | Bool | false | Cursor halo overlay |
| ShowCursorPulse | Bool | false | Cursor finder pulse |

| | | | |
|---|---|---|---|
| ShowFocusRing | Bool | false | UIA focus ring overlay |
| TtsZoomAnnounce | Bool | false | Speak zoom level on change |
| UiScale | Enum | 1.0 | 1.0/1.25/1.5/2.0 |
| ThemeName | String | Dark | One of 20 themes |

# 13. Themes

All 20 themes available. Control panel respects the active theme; the lens window renders screen content unaffected by theme colours.

# 14. Accessibility

| Criterion | Level | Status |
|---|---|---|
| 1.4.6 Contrast | AAA | Control panel ≥7:1 on all themes |
| 2.5.5 Target Size | AA | All controls ≥44×44 px |
| 2.4.7 Focus Visible | AA | 2 px Gold/Cyan focus ring |
| 1.4.1 Use of Colour | A | Filter names shown as text, not colour swatches alone |
| 2.1.1 Keyboard | A | All zoom, filter, overlay ops accessible via hotkeys |

# 15. Tips for Gaming

- Use Fixed mode with a corner-pinned lens for minimap detail without distracting lens drift.
- Enable HideFromCapture in Settings so the lens doesn't appear in stream footage.
- Assign per-app presets for each game executable — automatic preset switching handles mode changes on alt-tab.
- Cursor Finder Pulse helps locate the cursor after fast mouse movements in high-zoom scenarios.
- Warm filter reduces eye strain during long sessions.

# 16. Troubleshooting

| Symptom | Fix |
|---|---|
| Lens shows black rectangle | GPU doesn't support DWM compositing — update drivers; try 'Basic' rendering mode |
| Caret follow doesn't work | Target app doesn't implement UIA TextPattern — switch to Cursor mode |
| High GPU usage at high zoom | Reduce lens size; the Magnification API scales work with lens area |
| Per-app preset not triggering | Process name must match exactly (no path, include .exe extension) |
| TTS not speaking | Check Windows TTS voice installed; test in Settings → Accessibility → TTS Test |

| | |
|---|---|
| Colour picker off by one pixel | Disable display scaling in Windows Settings → Display → Scale |

# 17. Build Notes

Version v2026.05.09.0000 built with .NET 10, C# 14, WPF. Magnification API via P/Invoke to magnification.dll. DWM composition via dwmapi.dll. Per-monitor DPI awareness via SetThreadDpiAwarenessContext. JSON source generation for settings. Versioning: yyyy.MM.dd.NNN.

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